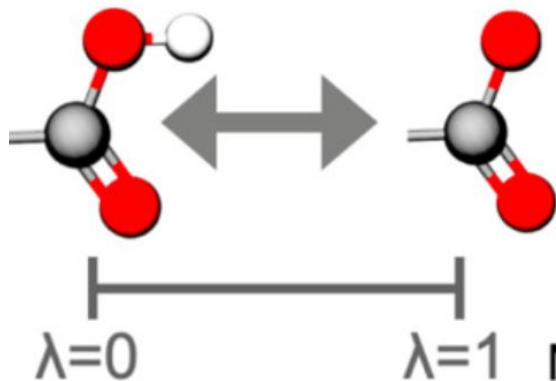


Constant pH MD simulations in AMBER



What is pH?

pH is a thermodynamic measure of proton chemical potential

- Defined as:

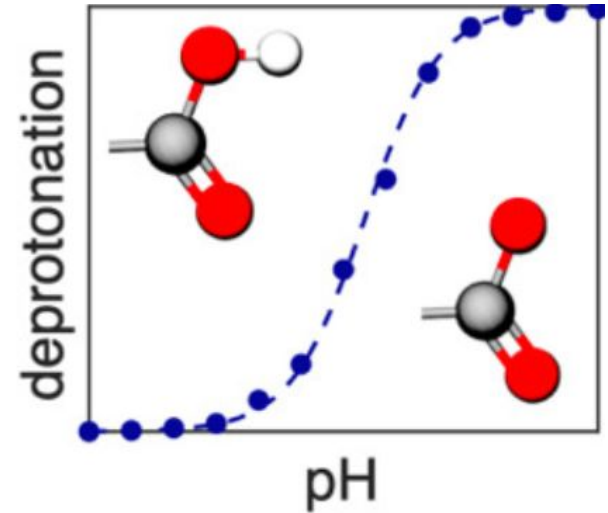
$$\text{pH} = -\log_{10} a(\text{H}^+)$$

- $a(\text{H}^+) = \mathbf{activity}$ of protons (not just concentration) but in practice we pretend it's the same.
- pH controls the **free energy cost** of adding/removing a proton
- free H^+ in solution do not really exist, it should be H_3O^+

Protonation is dynamic

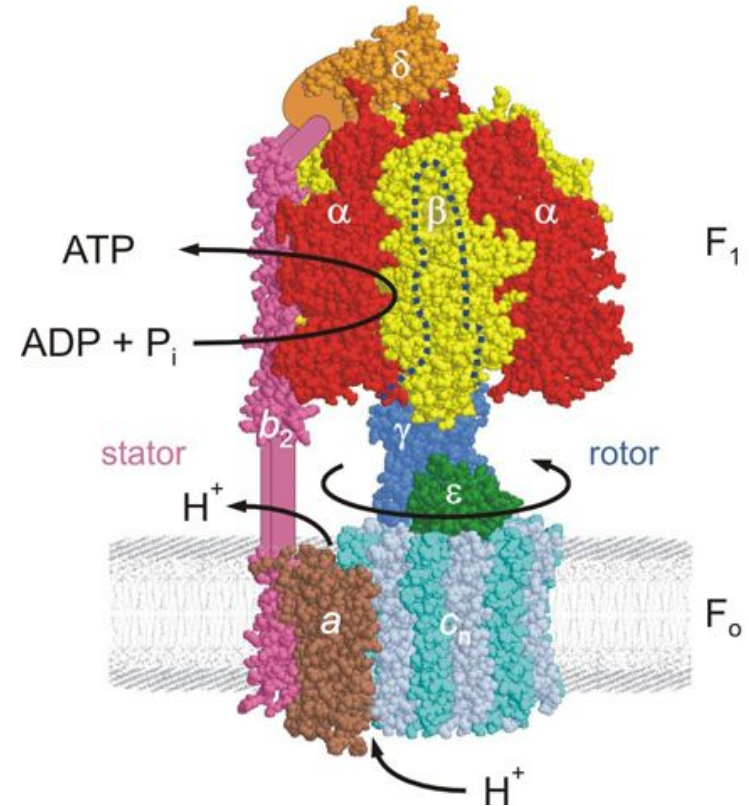
- Protonation is a thermodynamic variable
- Depends on pH and environment
- Controls charge and electrostatics

pKa: the pH value at which half of the titratable atoms are protonated.



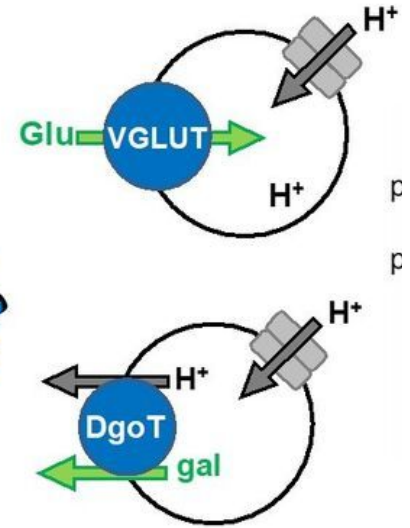
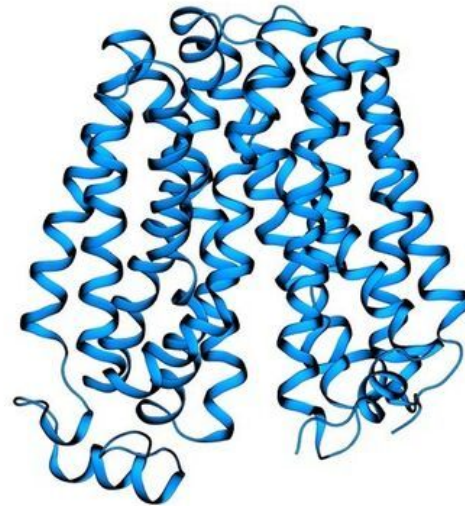
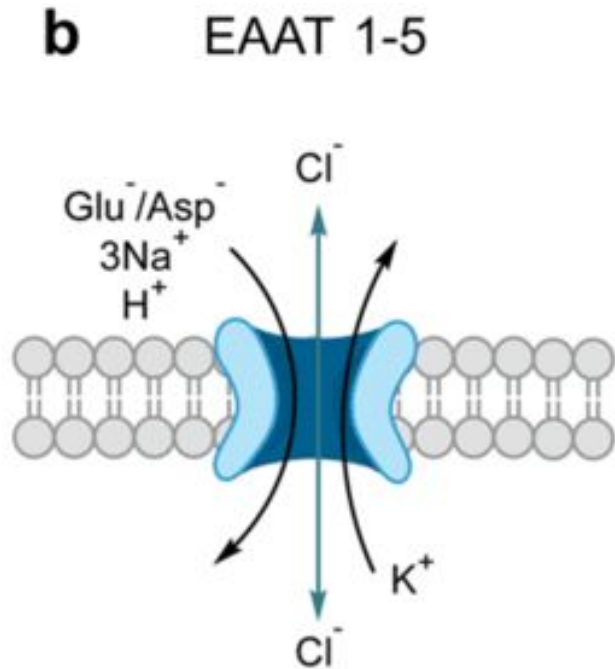
Why Protonation Matters in Biology

- Enzyme catalysis
- pH-dependent binding
- pH-dependent conformational changes
- Proton transport
- Protein stability
- secondary active transport



Many transporters are coupled to H^+ gradients

MFS superfamily has many examples



Model Compound pKa values for amino acids

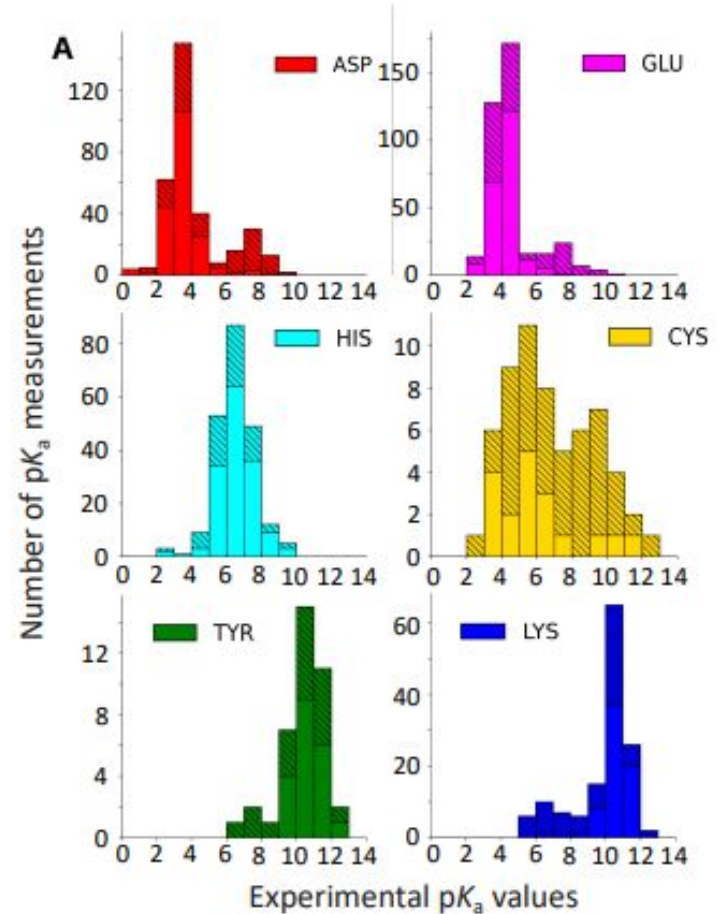
pKa of pure AA in solution

- Asp ~ 4.0
- Glu ~ 4.4
- His ~ 6.5
- Lys ~ 10.4
- Tyr ~ 10.1
- Cys ~ 8.3

Large pKa shifts are common

pKa of pure AA in solution

- Asp ~ 4.0
- Glu ~ 4.4
- His ~ 6.5
- Lys ~ 10.4
- Tyr ~ 10.1
- Cys ~ 8.3



What causes pKa shifts?

- Electrostatic environment
- Solvent exposure
- Hydrogen bonding
- Nearby charges

Standard MD with fixed protonation

- Protonation states assigned once
- Charges remain fixed
- No such thing as a pH

Implicit assumptions for fixed protonation

1. Protonation equilibrates instantly
2. One dominant protonation state
3. pKa values are environment-independent

Problems

- pKa values change in proteins
- Can shift by several units
- Could be wrong even without any changes pKa

Problems

- Wrong charge distribution
- Alters stability and interactions
- Protonation and conformation are coupled
- Protonation changes structure
- Structure changes pKa
- Fixed MD samples only one branch

Assigning protonation states

1. Run starting structure through PROPKA to get pKa estimate
2. Assign protonation states based on pKa and the pH the simulations are supposed to represent

Assigning protonation states

PROPKA is the most used

Based on empirical function of local environment

Nowadays there are many alternatives based on ML

KaMLs: Structure based tree model

Takes PDB as input

Calculates structure based features describing local environment

Feed those in a tree model

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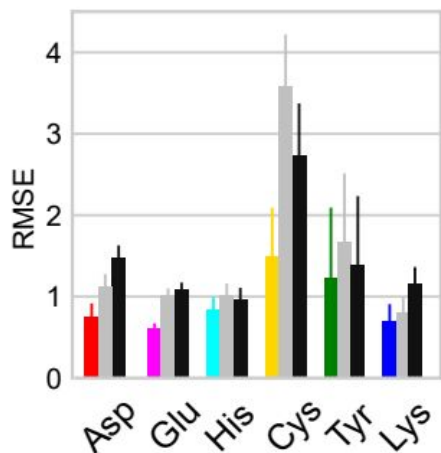
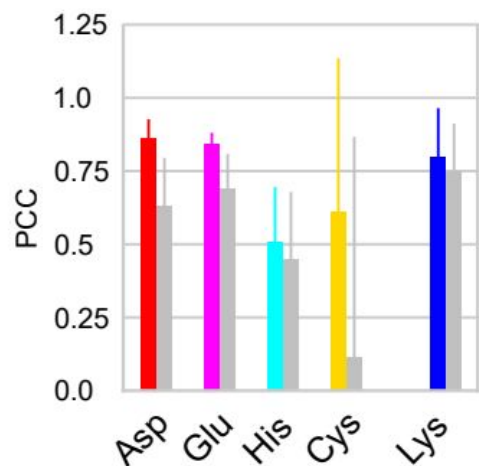
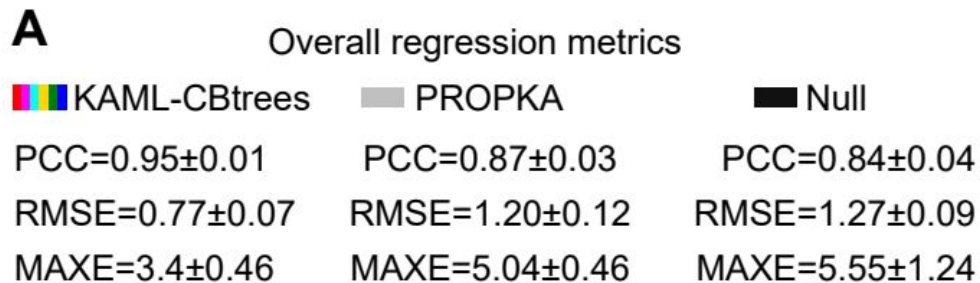
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BIOMOLECULAR SYSTEMS | January 30, 2025

KaMLs for Predicting Protein pK_a Values and Ionization States: Are Trees All You Need?

Mingzhe Shen, Daniel Kortzak, Simon Ambrozak, Shubham Bhatnagar, Ian Buchanan, Ruibin Liu, and Jana Shen*

KaMLs: Structure based tree model



PROPKA almost as bad as just taking model pKa of each amino acid

For certain residues even worse

KaML tree model outperforms PROPKA for each amino acid

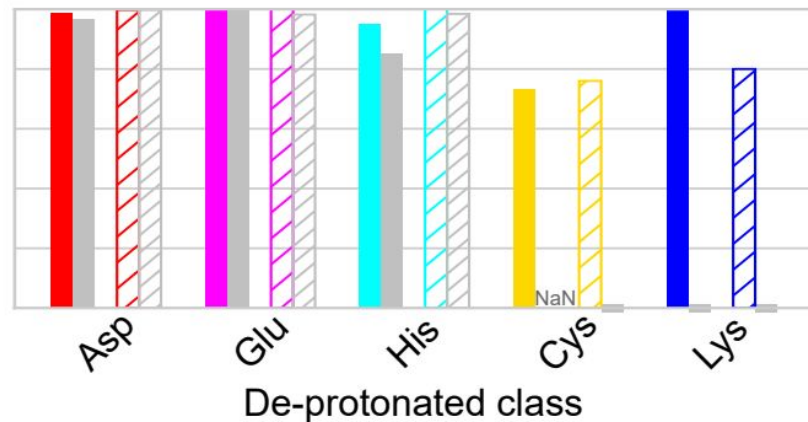
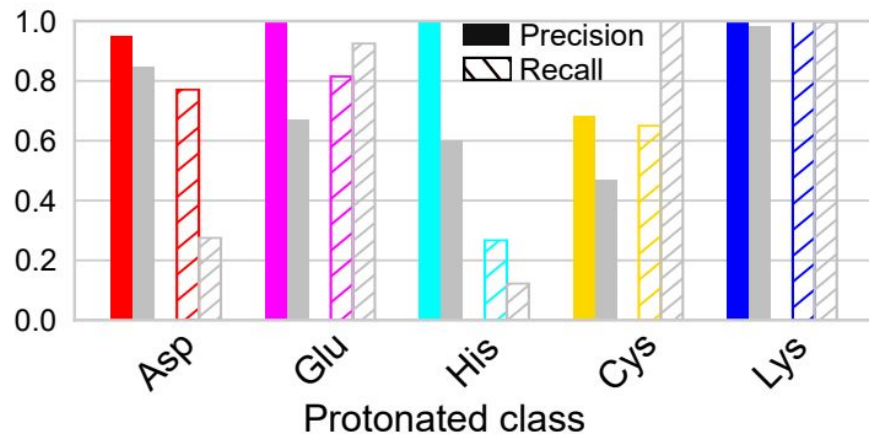
Assigning protonation states given neutral pH

Previous results are for pKa prediction

Our use case: Assign protonation states at a given pH

We can also compare how pKa predictors perform in this classification task

Assigning protonation states given neutral pH



PROPKA is particularly bad for the interesting cases: it almost never predicts Asp or His to be protonated.

Don't use PROPKA

There are many other alternatives

Every single one of them
outperforms PROPKA

There will also be an updated
version of KaMLs

Method	RMSE	PCC	R2
PypKa ⁹	0.82 ^a	na	na
PKAI ¹⁷	1.15 ^a	na	0.82
PKAI+ ¹⁷	0.98 ^a	na	0.87
DeepKa ^{14,15}	1.05 ^{a,b}	0.93 ^{a,b}	na
ANI-2X ⁵⁶			
KaML-CBtrees	0.77	0.95	0.88

^a as published by the authors for their own test sets. ^b CYS and TYR not included

Protein Electrostatic Properties are Fine-Tuned Through Evolution

 Mingzhe Shen,  Guy W. Dayhoff II,  Jana Shen

doi: <https://doi.org/10.1101/2025.04.17.649309>

This article is a preprint and has not been certified by peer review [what does this mean?].

History of constant pH MD

> [Proteins](#). 1997 Apr;27(4):523-44.

Simulation of protein conformational freedom as a function of pH: constant-pH molecular dynamics using implicit titration

A M Baptista ¹, P J Martel, S B Petersen

- **Baptista, Martel, and Petersen (1997)**

One of the first practical CpH-MD schemes, combining:

- MD for conformational sampling
- Monte Carlo (MC) moves to change protonation states
- Acceptance governed by pH and model compound pK_a values

This established the **MD–MC hybrid paradigm**, which remains influential.

History of constant pH MD

Typical workflow:

1. Run MD for a short time with fixed protonation
2. Attempt protonation/deprotonation moves via MC
3. Accept/reject based on:
 - Free-energy change
 - Solution pH
 - Reference pK_a values

Limitations of discrete CpHMD

- Discontinuous forces
- Sampling inefficiency
- Poor scaling

Continuous CpHMD

Constant-pH molecular dynamics using continuous titration coordinates

Michael S Lee¹, Freddie R Salsbury Jr, Charles L Brooks 3rd

- Introducing a continuous variable (λ) interpolating between protonated and deprotonated states
- Treating λ as a dynamical degree of freedom
- Protonation states emerge naturally as $\lambda \rightarrow 0$ or 1
- only implicit solvent

Continuous CpHMD

Advantages

- No discrete jumps in charge
- Better energy conservation
- Naturally coupled protonation and conformational dynamics

Challenges

- Requires careful biasing potentials
- Sampling efficiency can still be limited
- Interpretation of intermediate λ states requires care

Continuous CpHMD

Latest advances

- Proton tautomer sampling
- Explicit solvent
- Replica-exchange in pH space (pH-REMD)

► [J Chem Theory Comput.](#) Author manuscript; available in PMC: 2019 Mar 20.

Published in final edited form as: J Chem Theory Comput. 2011 Jul 1;7(8):2617–2629. doi: [10.1021/ct200146j](https://doi.org/10.1021/ct200146j) [↗](#)

Continuous Constant pH Molecular Dynamics in Explicit Solvent with pH-Based Replica Exchange

[Jason A Wallace](#)¹, [Jana K Shen](#)^{1,*}

λ -Dynamics vs Alchemical λ

CpHMD λ

- physical coordinate or a particle (even has a mass)
- Value of λ changed by forces acting on it

Alchemical λ (FEP/TI)

- Used for free-energy calculations
- Not a physical coordinate just a parameter
- Value of lambda is defined externally

Implementation

This one is used successfully, however its GROMACS 3.x

Donnini, S.; Tegeler, F.; Groenhof, G.; Grubmüller, H.: Constant pH molecular dynamics in explicit solvent with lambda-dynamics. *Journal of Chemical Theory and Computation* **7** (6), pp. 1962 - 1978 (2011)

[*Journal of Chemical Theory and Computation*](#) > Vol 21/Issue 4 > Article

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MOLECULAR MECHANICS | February 7, 2025

Constant pH Simulation with FMM Electrostatics in GROMACS. (B) GPU Accelerated Hamiltonian Interpolation

Bartosz Kohnke, Eliane Briand, Carsten Kutzner, and Helmut Grubmüller*

Implementation

Journal of Chemical Theory and Computation > Vol 18/Issue 10 > Article

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MOLECULAR MECHANICS | September 21, 2022

Scalable Constant pH Molecular Dynamics in GROMACS

Noora Aho*, Pavel Buslaev*, Anton Jansen, Paul Bauer, Gerrit Groenhof*, and Berk Hess*

KNOWN ISSUES

- Currently, separate PME ranks are not supported with MPI
- Code is not sufficiently covered with tests and some standard GROMACS test might be broken. Thus, `make check` might end up with errors. You can just ignore them.
- `cphmd` command currently does not work with `.edr` files saved as separate part after continuation (when `-noappend` is used in `mdrun`). We currently recommend all the users to append data to the initial files when continuing simulations.

I told this to them 3 years ago, still not fixed and what they call a solution was not an option for me...

Implementation

Journal of Chemical Theory and Computation > Vol 18/Issue 10 > Article

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MOLECULAR MECHANICS | September 21, 2022

Scalable Constant pH Molecular Dynamics in GROMACS

Noora Aho*, Pavel Buslaev*, Anton Jansen, Paul Bauer, Gerrit Groenhof*, and Berk Hess*

More problematic: In my hands simulated and experimental pKas were off by 2 units

Implementation for this course

Journal of Chemical Theory and Computation > Vol 18/Issue 12 > Article



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MOLECULAR MECHANICS | November 15, 2022

GPU-Accelerated All-Atom Particle-Mesh Ewald Continuous Constant pH Molecular Dynamics in Amber

Julie A. Harris, Ruibin Liu, Vinicius Martins de Oliveira, Erik Antonio Vázquez-Montelongo, Jack A. Henderson, and Jana Shen*

Implementation

Add protonation state as a “particle” with a mass and forces acting on it in the simulation system

$$\mathcal{H}(\{\mathbf{r}_i\}, \{\lambda_\alpha\}) = \frac{1}{2} \sum_i m_i \dot{\mathbf{r}}_i^2 + \frac{1}{2} \sum_\alpha \boxed{m_\alpha} \dot{\lambda}_\alpha^2 + U^{\text{ind}}(\{\mathbf{r}_i\}) \\ + U^{\text{hybr}}(\{\mathbf{r}_i\}, \{\lambda_\alpha\}) + \sum_\alpha U^*(\lambda_\alpha),$$

$$U^*(\lambda_\alpha) = -U^{\text{mod}}(\lambda_\alpha) + U^{\text{barr}}(\lambda_\alpha) + U^{\text{pH}}(\lambda_\alpha).$$

Implementation

U^{hybr}

All lambda dependent interactions (LJ and electrostatics of titratable side-chains)

U^{mod}

PMF of protonation for amino-acid in solution (simulated not measured, stored as coefficients of polynomial fit)

Implementation

U^{hybr} How much the system likes a proton at a certain position

U^{mod} Intrinsic affinity of a proton of side-chains

Also the only component which is residue specific, i.e. only need to simulate this to add new residues to CpHMD simulations

Implementation

$$U^{\text{barr}}(\lambda_\alpha) = 4\beta(\lambda_\alpha - 1/2)^2$$

Barrier potential preventing sampling of intermediate values

$$U^{\text{pH}}(\lambda_\alpha) = \ln 10 \cdot k_{\text{B}} T (\text{pH} - pK_a^{\text{mod}}) \lambda_\alpha$$

Solution pH goes in here.

Implementation

$$U^{\text{barr}}(\lambda_\alpha) = 4\beta(\lambda_\alpha - 1/2)^2$$

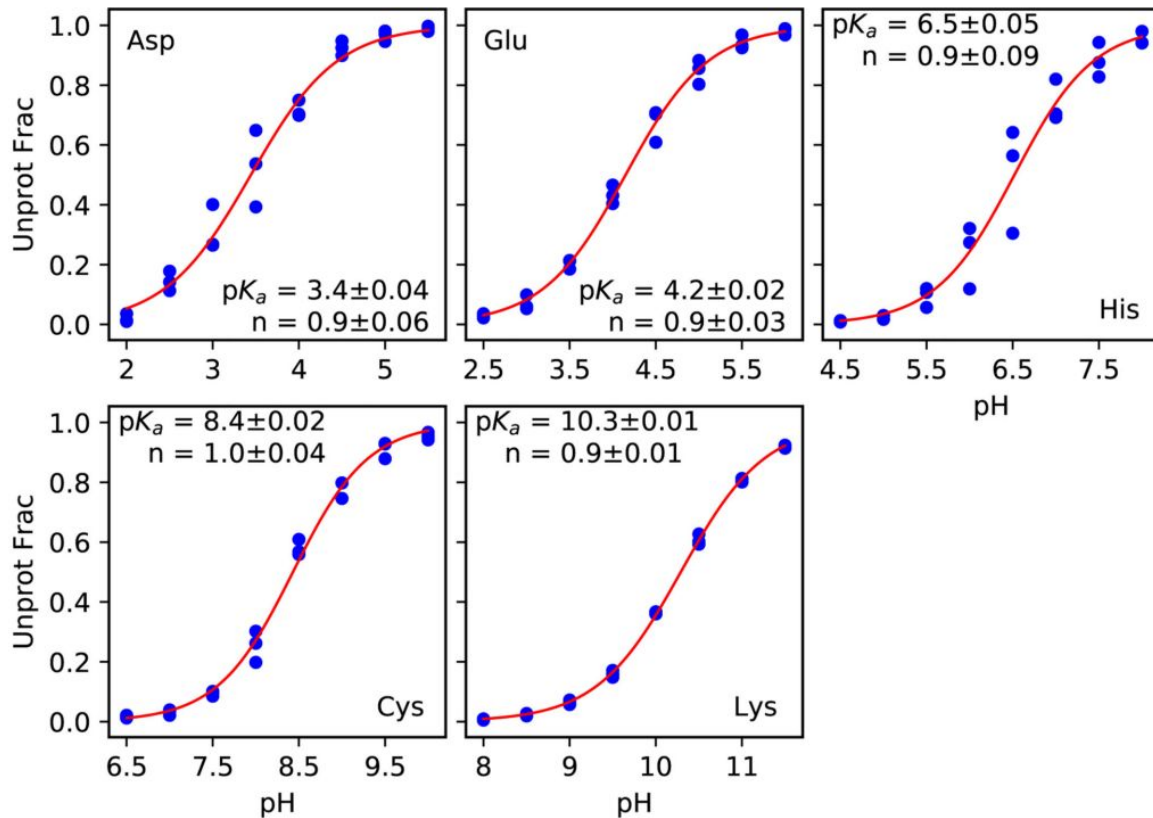
Barrier potential preventing sampling of intermediate values

$$U^{\text{pH}}(\lambda_\alpha) = \ln 10 \cdot k_B T (\text{pH} - pK_a^{\text{mod}}) \lambda_\alpha$$

Solution pH goes in here.

Also depends on protonation state. pH is irrelevant for a already protonated residue

Implementation



pKa of pure AA in solution

- Asp ~ 4.0
- Glu ~ 4.4
- His ~ 6.5
- Lys ~ 10.4
- Tyr ~ 10.1
- Cys ~ 8.3

Notes on the barrier potential

The barrier potential preventing sampling of intermediate values is chosen by us and might be adjusted in different systems

The rate of protonation changes depends on this barrier, i.e. they should be interpreted with care

They can be used to compare titration rates for e.g. the same protein in different conformations but their absolute values are meaningless

Only the probability to be protonated should be compared to experimental values

Prerequisites for the practical part

Software:

charmm, amber, parmed

system to simulate

Hardware:

Computer with nvidia GPU